

ASSIGNMENT 5:

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CLASS : SY BTECH IT

ABSTRACT CLASSES

/* 5) Implement a program using an abstract class named Shape to calculate the area and perimeter of different geometric shapes. Derive the classes Circle, Square, Rectangle, and Triangle from the Shape class. Each derived class should override the abstract methods to calculate the area and perimeter according to the respective shape's formula.

*/

Import java.util.*;

// Abstract Class - a class which has at least one method without body

// Abstract class needs to be inherited and objects of such classes cannot be created

```
abstract class Shape {  
    abstract void area();  
    abstract void perimeter();  
}
```

```
class Circle extends Shape {  
    double r;  
    Circle(double r) {  
        this.r = r;  
    }  
    // Method to calculate Area of Circle  
    void area() {  
        double a = 3.14*r*r;  
        System.out.println("Area Of Circle is " + a);  
    }  
    // Method to calculate Circumference of Circle  
    void perimeter() {  
        double per = 2*3.14*r;  
        System.out.println("Circumference of Circle is " + per);  
    }  
}
```

```
}
```

```
class Square extends Shape {  
    double s;  
    Square(double s) {  
        this.s = s;  
    }  
    // Method to calculate Area of Square  
    void area() {  
        double a = s*s;  
        System.out.println("Area Of Square is " + a);  
    }  
    // Method to calculate Perimeter of Square  
    void perimeter() {  
        double per = 4*s;  
        System.out.println("Perimeter of Square is " + per);  
    }  
}
```

```
class Rectangle extends Shape {  
    double l, b;  
    Rectangle(double l, double b) {  
        this.l = l;  
        this.b = b;  
    }  
    // Method to calculate Area of Rectangle  
    void area() {  
        double a = l*b;  
        System.out.println("Area Of Rectangle is " + a);  
    }  
    // Method to calculate Perimeter of Rectangle  
    void perimeter() {  
        double per = 2*(l+b);  
        System.out.println("Perimeter of Rectangle is " + per);  
    }  
}
```

```
class Triangle extends Shape {
```

```

double a, b, c;
Triangle(double a, double b, double c) {
    this.a = a;
    this.b = b;
    this.c = c;
}
// Method to calculate Area of Triangle
void area() {
    double s = (a + b + c) / 2;
    double ar = Math.sqrt(s * (s - a) * (s - b) * (s - c));
    System.out.println("Area Of Triangle is " + ar);
}
// Method to calculate Perimeter of Triangle
void perimeter() {
    double per = a + b + c;
    System.out.println("Perimeter of Triangle is " + per);
}
}

public class Main {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        int ch;
        do {
            System.out.println("\n***** MENU *****");
            System.out.println("1. Circle");
            System.out.println("2. Square");
            System.out.println("3. Rectangle");
            System.out.println("4. Triangle");
            System.out.println("5. Exit");
            System.out.print("Enter your choice: ");
            ch = sc.nextInt();
            switch (ch) {
                case 1:
                    System.out.print("Enter the radius of circle: ");
                    double r = sc.nextDouble();
                    Circle cir = new Circle(r);
                    cir.area();

```

```

        cir.perimeter();
        break;
    case 2:
        System.out.print("Enter the side of square: ");
        double s = sc.nextDouble();
        Square sq = new Square(s);
        sq.area();
        sq.perimeter();
        break;
    case 3:
        System.out.print("Enter the length and breadth of rectangle: ");
        double l = sc.nextDouble();
        double b = sc.nextDouble();
        Rectangle rect = new Rectangle(l, b);
        rect.area();
        rect.perimeter();
        break;
    case 4:
        System.out.print("Enter the three sides of triangle: ");
        double a = sc.nextDouble();
        double b1 = sc.nextDouble();
        double c = sc.nextDouble();
        Triangle tri = new Triangle(a, b1, c);
        tri.area();
        tri.perimeter();
        break;
    case 5:
        System.out.println("Successfully Exited");
        break;
    default:
        System.out.println("Invalid Choice");
    }
} while (ch != 5);
}
}

```

OUTPUT:

```
Command Prompt
C:\Users\DIYA\Desktop>java Main

**** MENU ****
1. Circle
2. Square
3. Rectangle
4. Triangle
5. Exit
Enter your choice: 1
Enter the radius of circle: 2
Area Of Circle is 12.56
Circumference of Circle is 12.56

**** MENU ****
1. Circle
2. Square
3. Rectangle
4. Triangle
5. Exit
Enter your choice: 2
Enter the side of square: 2
Area Of Square is 4.0
Perimeter of Square is 8.0

**** MENU ****
1. Circle
```

```
Command Prompt
2. Square
3. Rectangle
4. Triangle
5. Exit
Enter your choice: 3
Enter the length and breadth of rectangle: 2
3
Area Of Rectangle is 6.0
Perimeter of Rectangle is 10.0

**** MENU ****
1. Circle
2. Square
3. Rectangle
4. Triangle
5. Exit
Enter your choice: 4
Enter the three sides of triangle: 3 4 5
Area Of Triangle is 6.0
Perimeter of Triangle is 12.0

**** MENU ****
1. Circle
2. Square
3. Rectangle
4. Triangle
5. Exit
Enter your choice: 5
Successfully Exited
```